Multi-Source Candidate Data Transformer — Technical Design

Dasoju Shashank | shashankdasoju111@gmail.com | Eightfold Engineering Intern Assignment

1. PIPELINE ARCHITECTURE

Stage	What happens	Resilience
Source Dialect Interpreters	Infer source type (CSV, JSON, PDF, DOCX, TXT, GitHub). Each extractor handles its own dialect.	Unknown → skip
Canonicalization Bus	Phones → E.164. Dates → YYYY-MM. Country → ISO-3166. Emails → lowercase. Skills → alias map.	Unparseable → None
Identity Clustering Layer	O(n) graph-based grouping by email/phone. Fallback to normalized name + country.	No match → single
Staged Consensus Engine	Scalar fields resolved by dynamic source quality. List fields union + dedupe. Experience entries merged with granular date selection.	Empty → zero weight
Trust & Corroboration Vault	Per-skill: freq×0.5 + avg_quality×0.5. Overall: coverage×0.5 + avg_quality×0.5. Quality is dynamic—based on completeness.	Bounded [0,1]
Projection Gateway	Runtime config selects/renames via path DSL (emails[0], skills[].name, location.city). Applies per-field normalization, on_missing, type validation.	Missing required → ProjectionError
Schema Vestibule	Post-projection type/required checks. Errors surfaced as list—pipeline never crashes. Supports source veto (ignored_sources).	Returns error list

2. CANONICAL OUTPUT SCHEMA

Field	Type	Normalization
candidate_id	UUIDv5	deterministic
full_name	str | None	—
emails	list[str]	lowercase, RFC-5322
phones	list[str]	E.164
location	{city, region, country}	country → ISO-3166
links	{linkedin, github, portfolio, other[]}	—
headline	str | None	—
years_experience	float | None	—
skills	[{name, confidence, sources[]}]	alias map → canonical
experience	[{company,title,start,end, summary}]	granular date merge
education	[{institution,degree,field, end_year}]	end_year → int
provenance	[{field, source, method}]	every accepted value
overall_confidence	float [0,1]	dynamic source scoring

3. MERGE & CONFLICT RESOLUTION

Identity Clustering: email > phone > name. O(n) graph-based.

Scalar fields (name, headline):

- Pick highest **dynamic quality score** (completeness, not file type)
- Tie → first; flagged as 'conflict-resolved'

List fields (emails, phones, skills):

- Union + normalize → dedupe
- Skills: confidence = freq×0.5 + avg_quality×0.5

Experience (Granular Merge):

- Groups by normalized company
- For duplicates, picks most precise date range
- Concatenates unique summaries

4. RUNTIME CONFIG & PROJECTION

Clean separation: engine → CanonicalProfile. Projection Gateway reshapes.

Config capabilities:

- **fields[].path** — output key
- **fields[].from** — source path (supports [0] and [].)
- **fields[].normalize** — E164 | canonical | lowercase
- **fields[].required** — drives on_missing
- **ignored_sources** — skip source types (veto)
- **on_missing**: 'null' | 'omit' | 'error'
- **include_confidence / include_provenance** toggles

Path DSL examples:

`emails[0]` → first email `skills[].name` → all `location.country` → nested

5. EDGE CASES & HANDLING

Edge case	How handled
Missing / empty source	Try/except; returns [].
Multiple candidates (CSV 3 rows → 3 profiles)	Identity Clustering groups by email/phone.
Name capitalisation conflict	Normalized lowercase for match; highest-quality for display.
Duplicate phone formats	Normalized to E.164; dict-key dedup.
Unknown ATS fields	Ignored (never invented).
Single-source skill	Confidence = source_quality × 0.5.
Ignore known-bad source	Runtime config supports 'ignored_sources'.

6. DELIBERATE DESCOPING

- LinkedIn scraping skipped (ToS); URL stored.
- OCR for scanned PDFs not included.
- Async/concurrent not implemented.
- Resume NER uses regex (fast, deterministic, zero hallucinations).

Implementation: Python 3.12 · pydantic · phonenumbers · pycountry · pypdf · python-docx · requests · reportlab | CLI: python cli.py --sources [--config] [--output] | Demo: ./run_demo.sh | Tests: pytest tests/ -v | Identity Clustering + Dynamic Scoring + Granular Merge